

Technical Cyber security Report

Analysis Using Nmap for Scanning Local IP

By

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Date: 9th April 2025

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REPORT OUTLINE

1. Executive Summary	3
1.1. Overview.....	3
1.2. Key Findings	3
2. Background and Objectives	3
2.1. Project Context	3
2.2. Objective of the Tool Use:.....	3
3. Methodology	3
3.1 Tool Configuration	3
3.2 Execution Process	4
3.3 Monitoring and Analysis	7
4. Findings and Analysis	7
4.1 Indicators of Compromise (IOCs).....	7
4.2 Behavioural Analysis.....	7
4.3 Risk and Impact Assessment	8
5. Recommendations	8
5.1 Immediate Remediation Actions:	8
5.2 Long-Term Mitigation	8
6. Conclusion.....	8
Appendix: Additional Data.....	8

1. Executive Summary

1.1. Overview

In order to uncover any potential vulnerabilities and open ports that may be present in the local IP address of 192.168.56.1, an analysis was carried out with the help of Nmap. The tool successfully gave insights into the IP addresses open ports, the names of the applications producing these open ports, and the operating system on which that IP address is running, allowing the user to take prompt action to remedy those open ports.

1.2. Key Findings

Nmap identifies there were 7 open ports after scanning 1000 ports on the local IP address 192.168.56.1. The open ports were flagged for further assessment on the network.

2. Background and Objectives

2.1. Project Context

The project aimed to assess the security posture of the network of the local IP address 192.168.56.1 and identify any vulnerable ports that could compromise the network.

2.2. Objective of the Tool Use:

Nmap was utilised for the purpose of finding open ports, detecting actively running services, and assessing firewall configurations of the local network, thereby improving security by allowing for proactive remedies such as port closure or software updates.

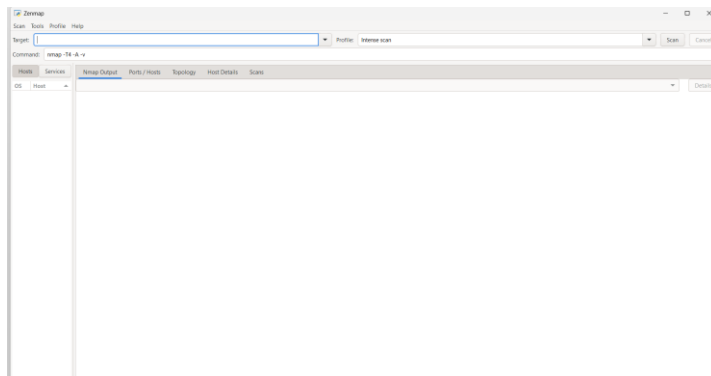
3. Methodology

3.1 Tool Configuration

Nmap Configuration:

1. Download the Nmap Installer
2. Run the Installer
3. Follow the on-screen instructions to complete the installation process.

4. To Verify the Installation, you open the Command Prompt (cmd).
5. Type nmap and press Enter. If Nmap runs and shows its usage options, the installation was successful.



3.2 Execution Process

File Execution in Nmap:

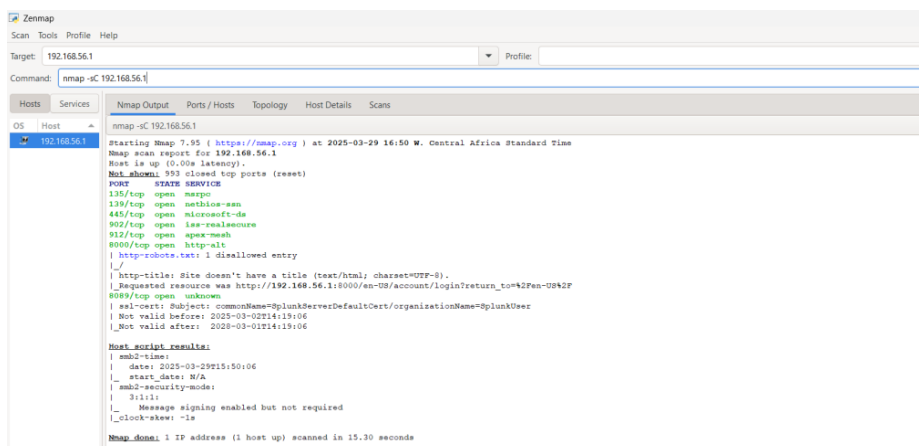
To fully execute the network scanning on Nmap using these commands **-sC**, **-sV**, **-vv**, **-sL**, **-sn**, and **-Pn**, here are the steps I followed:

- **Network Scanning using the -sC**

Step 1: Open command prompt

Step 2: Enter ipconfig to get IP Address of your system – 192.168.56.1

Step 3: Open Nmap and enter the IP address in the target port and change the command to -sC

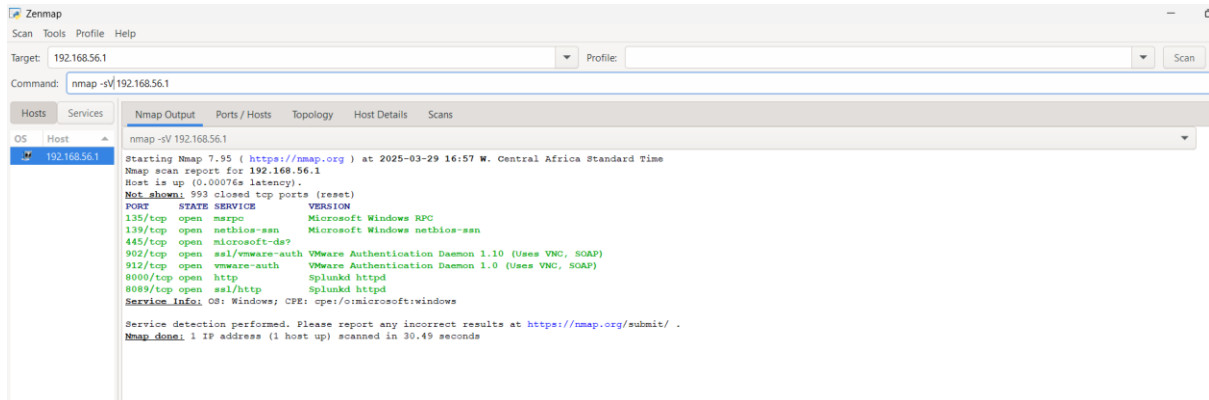


The Results

- The host is shown as up and running with a latency value of 0.0s
- Showing that 993 ports are closed and 7 ports are opened.
- Open ports are: 135/tcp ,139/tcp, 445/tcp, 902/tcp, 912/tcp, 8000/tcp, 8089/tcp

- The total time for the scan was 15.30 seconds
- **Network Scanning using the -sV**

Step 6: Change the command to -sV

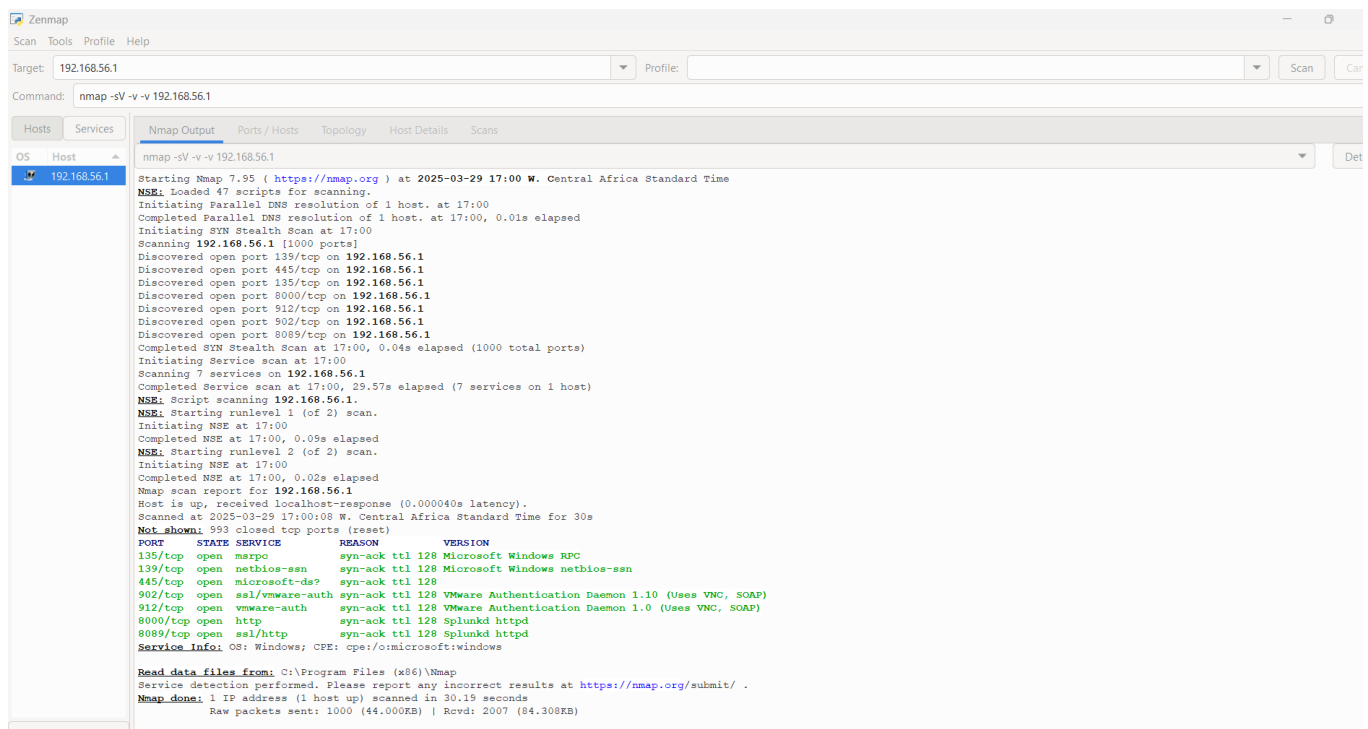


Result

- The host is shown as up and running with a latency value of 0.00076s
- There are 993 closed ports
- Open ports are: 135/tcp ,139/tcp, 445/tcp, 902/tcp, 912/tcp, 8000/tcp, 8089/tcp
- The version of the system is visible: OS: Windows; CPE: cpe:/o:microsoft:windows
- The total time for the scan was 30.49 seconds

- **Network Scanning using the -vv**

Step 7: Add -vv to the -sV command

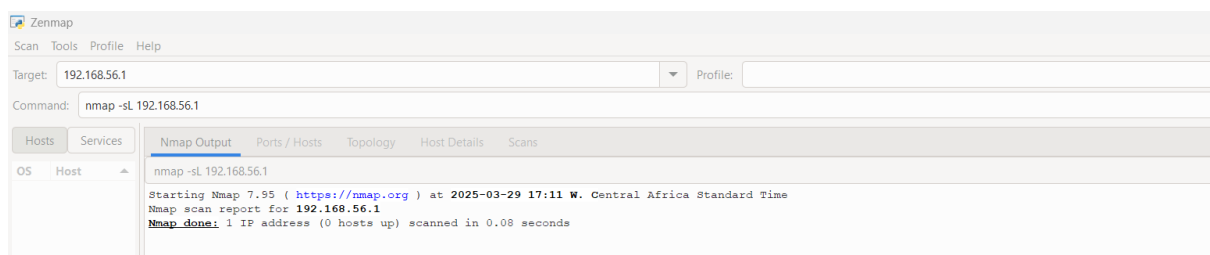


Result (A lot more detailed)

- 47 scripts were loaded for scanning 1000 ports
- The time taken to run the SYN Stealth scan was 0.04s
- Open ports are: 135/tcp ,139/tcp, 445/tcp, 902/tcp, 912/tcp, 8000/tcp, 8089/tcp
- All the windows version for the open ports were highlighted.

• Network Scanning using the -sL

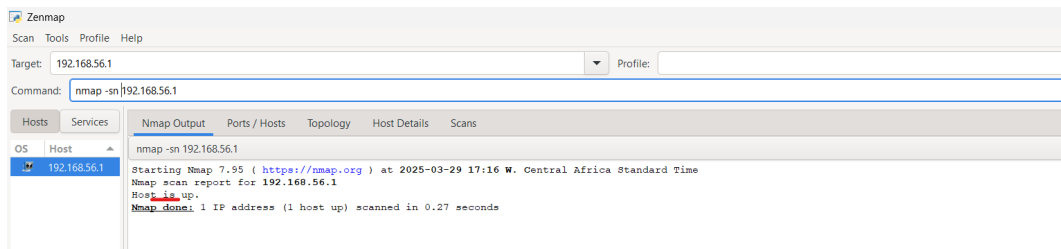
Step 8: Change the command to -sL,



Result: A summarised result showing that 0 host is up and it was scanned on 0.06 seconds

• Network Scanning using the -sn

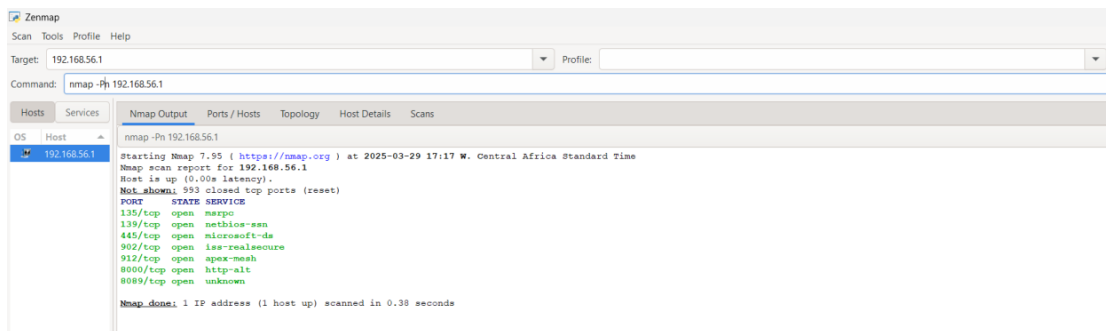
Step 9: Change the command to -sn



Result: This shows that the host is up and the scanning was completed in 0.27seconds

- **Network Scanning using the -Pn**

Step 10: Change the command to -Pn



Result: This shows a summary of all the closed ports.

- Shows a list of the open ports and their service names

3.3 Monitoring and Analysis

- **System Monitoring:**
 - The Nmap recorded that 7 ports were opened in the Windows operating system due to the VM ware, Splunk and the Microsoft RPC that are all installed on the local system.
- **Network Monitoring:**

The system did Has a latency of less that 1 which meant that it was accessible. The scan did reveal that there were 7 open ports.

4. Findings and Analysis

4.1 Indicators of Compromise (IOCs)

- **Detected IOCs:**

There was no detected IOC in the result of the scanning.

4.2 Behavioural Analysis

- **Malware Actions:**

There was no Malware identified during the scan.

- **Persistence Mechanisms:**
Not applicable to the above scan.

4.3 Risk and Impact Assessment

- **Potential Impact:**
If exposed to malicious actors, open ports on local IP addresses can expose systems to vulnerabilities, unauthorized access, and increased exposure to attacks, potentially leading to system compromise, data breaches, or service disruption.
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5. Recommendations

5.1 Immediate Remediation Actions:

- **Closure of unused Ports**

To reduce hazards, all unused open ports should be closed

- **Use of Firewalls:**

Firewalls and Antivirus should be used in the local systems to restrict access to only trusted networks.

5.2 Long-Term Mitigation

- **Monitoring and Detection Enhancements:**
Recommending a constant monitoring of the IP address using Nmap to detect any unusual activities and open ports
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6. Conclusion

- **Summary of Findings:**
Nmap successfully identified all the open ports (7) out of the 1000 ports scanned in the Local IP address 192.168.56.1.
 - **Next Steps:**
Firewalls and Antivirus should be used immediately on the local systems to restrict access to only trusted networks.
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Appendix: Additional Data

- **Raw Logs and Tool Outputs:**
See the raw output from Nmap network monitoring tool

Zenmap

Scan Tools Profile Help

Target: 192.168.56.1 Profile:

Command: nmap -sC 192.168.56.1

Hosts Services

Nmap Output Ports / Hosts Topology Host Details Scans

OS Host

192.168.56.1

Starting Nmap 7.95 (<https://nmap.org>) at 2025-03-29 16:50 W. Central Africa Standard Time

Nmap scan report for 192.168.56.1

Host is up (0.00s latency).

Not shown: 993 closed top ports (reset)

PORT	STATE	SERVICE
135/tcp	open	msrpc
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds
902/tcp	open	lss-rpc
912/tcp	open	apex-mesh
8000/tcp	open	http-alt

| <http://robots.txt>: 1 disallowed entry

|/|

|_ http-title: Site doesn't have a title (text/html; charset=UTF-8).

|_ Requested resource was http://192.168.56.1:8000/en-09/account/login?return_to=42Pen-0242F

|_ 8000/tcp open unknown

|_ ssl-cert: Subject: commonName=SplunkdServerDefaultCert/organizationName=SplunkUser

|_ Not valid before: 2025-03-02T14:19:06

|_ Not valid after: 2028-03-02T14:19:06

Host script results:

|_ smb2-time:

|_ date: 2025-03-29T15:50:06

|_ start_date: N/A

|_ smb2-security-mode:

|_ 3111:

|_ Message signing enabled but not required

|_ clock-skew: -1s

Nmap done: 1 IP address (1 host up) scanned in 15.30 seconds

Zenmap

Scan Tools Profile Help

Target: 192.168.56.1 Profile:

Command: nmap -sV 192.168.56.1

Hosts Services

Nmap Output Ports / Hosts Topology Host Details Scans

OS Host

192.168.56.1

Starting Nmap 7.95 (<https://nmap.org>) at 2025-03-29 16:57 W. Central Africa Standard Time

Nmap scan report for 192.168.56.1

Host is up (0.00076s latency).

Not shown: 993 closed top ports (reset)

PORT	STATE	SERVICE	VERSION
135/tcp	open	msrpc	Microsoft Windows RPC
139/tcp	open	netbios-ssn	Microsoft Windows netbios-ssn
445/tcp	open	microsoft-ds	
902/tcp	open	ssl/vmware-auth	VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp	open	vmware-auth	VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
8000/tcp	open	http	Splunkd httpd
8089/tcp	open	ssl/http	Splunkd httpd

Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 30.49 seconds

Zenmap

Scan Tools Profile Help

Target: 192.168.56.1 Profile:

Command: nmap -sL 192.168.56.1

Hosts Services

Nmap Output Ports / Hosts Topology Host Details Scans

OS Host

192.168.56.1

Starting Nmap 7.95 (<https://nmap.org>) at 2025-03-29 17:11 W. Central Africa Standard Time

Nmap scan report for 192.168.56.1

Nmap done: 1 IP address (0 hosts up) scanned in 0.08 seconds

Zenmap

Scan Tools Profile Help

Target: 192.168.56.1 Profile:

Command: nmap -sV -v -v 192.168.56.1

Hosts Services

OS Host

192.168.56.1

Nmap Output

nmap -sV -v -v 192.168.56.1

Starting Nmap 7.95 (<https://nmap.org>) at 2025-03-29 17:00 W. Central Africa Standard Time

NSE: Loaded 47 scripts for scanning.

Initiating Parallel DNS resolution of 1 host. at 17:00

Completed Parallel DNS resolution of 1 host. at 17:00, 0.01s elapsed

Initiating SYN Stealth Scan at 17:00

Scanning 192.168.56.1 [1000 ports]

Discovered open port 135/tcp on 192.168.56.1

Discovered open port 445/tcp on 192.168.56.1

Discovered open port 135/tcp on 192.168.56.1

Discovered open port 8000/tcp on 192.168.56.1

Discovered open port 912/tcp on 192.168.56.1

Discovered open port 902/tcp on 192.168.56.1

Discovered open port 8089/tcp on 192.168.56.1

Completed SYN Stealth Scan at 17:00, 0.04s elapsed (1000 total ports)

Initiating Service scan at 17:00

Scanning 7 services on 192.168.56.1

Completed Service scan at 17:00, 29.57s elapsed (7 services on 1 host)

NSE: Script scanning 192.168.56.1.

NSE: Starting runlevel 1 (of 2) scan.

Initiating NSE at 17:00

Completed NSE at 17:00, 0.09s elapsed

NSE: Starting runlevel 2 (of 2) scan.

Initiating NSE at 17:00

Completed NSE at 17:00, 0.02s elapsed

Nmap scan report for 192.168.56.1

Host is up, received localhost-response (0.000040s latency).

Scanned at 2025-03-29 17:00:08 W. Central Africa Standard Time for 30s

Not shown: 993 closed tcp ports (reset)

PORT	STATE	SERVICE	REASON	VERSION
135/tcp	open	msrpe	syn-ack ttl 128	Microsoft Windows RPC
139/tcp	open	netbios-ssn	syn-ack ttl 128	Microsoft Windows netbios-ssn
445/tcp	open	microsoft-ds?	syn-ack ttl 128	
902/tcp	open	ssl/vmware-auth	syn-ack ttl 128	VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp	open	vmware-auth	syn-ack ttl 128	VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
8000/tcp	open	http	syn-ack ttl 128	Splunkd httpd
8089/tcp	open	ssl/http	syn-ack ttl 128	Splunkd httpd

Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Read data files from: C:\Program Files (x86)\Nmap

Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

Nmap done: 1 IP address (1 host up) scanned in 30.19 seconds

Raw packets sent: 1000 (44.000KB) | Rcvd: 2007 (84.308KB)

Zenmap

Scan Tools Profile Help

Target: 192.168.56.1 Profile:

Command: nmap -sn 192.168.56.1

Hosts Services

OS Host

192.168.56.1

Nmap Output

nmap -sn 192.168.56.1

Starting Nmap 7.95 (<https://nmap.org>) at 2025-03-29 17:16 W. Central Africa Standard Time

Nmap scan report for 192.168.56.1

Host is up.

Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds

Zenmap

Scan Tools Profile Help

Target: 192.168.56.1 Profile:

Command: nmap -Pn 192.168.56.1

Hosts Services

OS Host

192.168.56.1

Nmap Output

nmap -Pn 192.168.56.1

Starting Nmap 7.95 (<https://nmap.org>) at 2025-03-29 17:17 W. Central Africa Standard Time

Nmap scan report for 192.168.56.1

Host is up (0.00s latency).

Not shown: 993 closed top ports (reset)

PORT	STATE	SERVICE
135/tcp	open	msrpe
139/tcp	open	netbios-ssn
445/tcp	open	microsoft-ds
902/tcp	open	iss-realsecure
912/tcp	open	apex-mesh
8000/tcp	open	http-alt
8089/tcp	open	unknown

Nmap done: 1 IP address (1 host up) scanned in 0.38 seconds